MILITARY SPECIFICATION

INDICATOR, AZIMUTH, T23

1. SCOPE

1.1 This specification covers one type of azimuth indicator known as Indicator, Azimuth, T23 for use with Local Control System M16A1 to indicate azimuth deflection.

2. APPLICABLE SPECIFICATIONS AND DRAWINGS

2.1 Specifications - The following specifications and specifications referenced on drawings in 2.2, of the issue in effect on the date of invitation for bids, form a part of this specification;

FEDERAL SPECIFICATIONS

NN-B-601	Boxes; Wood-Cleated-Plywood
NN-B-621	Boxes; Wood, Nailed and Lock-Corner
NN-B-631	Boxes; Wood, Wirebound, for Domestic Shipment

MILITARY SPECIFICATIONS

JAN-P-100	Packaging and Packing for Overseas Shipment - General Specification
JAN-P-105	Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Plywood
JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed
JAN-P-108	Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board) Exterior and Interior
JAN-P-116	Packaging and Packing for Overseas Shipment - Preservation, Methods of
JAN-B-121	Barrier Materials, Greaseproof
JAN-P-125	Packaging and Packing for Overseas Shipment - Barrier Materials, Waterproof, Flexible
JAN-P-127	Packaging and Packing for Overseas Shipment - Tape, Adhesive, Pressure-Sensitive, Water-Resistant
JAN-P-140	Packaging and Packing for Overseas Shipment - Adhesives, Water-Resistant, Case Liner

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U. S. ARMY SPECIFICATIONS

2-134	Grease; Lubricating, Instrument
51-7	Instruments, Fire-Control; General Specification Governing the Manufacture, and Inspection of
100-2	Standard Specification for Marking Shipments by Contractors

(Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number should be stipulated when requesting copies.)

2.2 Drawings - The following drawings and all drawings referenced thereon, of the issue in effect on the date of invitation for bids, form a part of this specification:

ARMY ORDNANCE DEPARTMENT DRAWINGS

F7647798	Indicator, Aximuth, T23
B7633839	Cleaning, Preserving, Packaging (Methods O, I and IA,) Packing and Marking of Fire Control Materiel Spare Parts for Domestic Shipment and Storage, and Export Shipment.

(Copies of drawings should be obtained from the procuring agency or as directed by that agency. Both the title and the identifying number should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 General requirements and associated tests common to all fire control instruments as contained in Specification 51-7 shall be mandatory in so far as they are applicable.

3.2 Materials shall be in accordance with the drawings, general specifications and material specifications forming part of this specification.

3.3 Operational checks - Clutches, shutters, resetting devices and detents shall function properly.

3.4 Error - The maximum overall error including backlash between the counters and input shaft shall not exceed 1.5 mils (counter reading).

3.5 Illumination - The figures shall be clearly visible when examined under darkroom conditions with 3 volts applied.

3.6 Sealing - The indicator shall not leak when 5 PSI air pressure is applied internally.

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3.7 Temperature - The indicator shall meet the requirements of Paragraphs 3.3, 3.4, and 3.5 throughout the temperature range of -65°F to +150°F, and shall not be damaged by exposure throughout the temperature range of -80°F to +160°F.

3.8 Lubrication - Grease which meets the requirements of Specification 2-134 shall be used for lubrication.

3.9 Workmanship - Workmanship shall be in accordance with Specification 51-7.

4. SAMPLING, INSPECTION AND TEST PROCEDURE

4.1 Unless otherwise specified, each azimuth indicator and spare part shall be inspected for conformance with this specification and as specified in Specification 51-7.

4.2 Error test - Each instrument shall be tested for error at not less than 10 different settings.

4.3 Sealing test - The pipe plug in the bottom of the housing shall be removed and clean dry air shall be applied at a pressure of 5 PSI. The indicator shall be immersed in water. Leakage will be indicated by the continuous formation of air bubbles. The pipe plug shall be replaced when the test is completed.

4.4 Temperature test - Twenty azimuth indicators, selected from each lot of 100 or fraction thereof contracted for, shall be exposed to an ambient temperature of minus 80°F for 5 hours and subsequently at minus 65° F for 2 hours, and then tested for the requirements of Paragraph 3.7 at this latter temperature. The temperature of the instrument shall be returned to normal room temperature. The instrument shall then be exposed to an ambient temperature of plus 160°F for 5 hours and subsequently at plus 150°F for 2 hours, and then tested for the requirements of paragraph 3.7 at this latter temperature. The instrument shall then be exposed to normal room temperature and tested for all of the requirements of this specification.

4.5 Rejections - Individual azimuth indicators failing to meet the inspection tests shall be rejected. When tests are specified on a quantity of azimuth indicators that are selected as representative of a lot and one or more fail to meet the requirements, the lot shall be rejected. Rejected azimuth indicators shall not be resubmitted for inspection without furnishing full particulars concerning previous rejections and measures taken to correct the defects.

5. PREPARATION FOR DELIVERY

5.1 Packaging

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5.1.1 Azimuth Indicator T23 shall be cleaned in accordance with the applicable requirements of Specification JAN-P-116.

5.1.2 The Cable and Lamp Assembly shall be removed and then unit packaged in accordance with Specification JAN-P-116, Method IA-8. The threaded hole from which the Lamp Assembly was removed shall be covered with pressure sensitive tape conforming with the requirements of Type I, II or III, Grade B of Specification JAN-P-127. The azimuth indicator (less cable and lamp assembly) shall be wrapped with greaseproof barrier material conforming with the requirements of Type II, Grade C, Class 2 of Specification JAN-B-121. The wrapped indicator and prepackaged cable and lamp assembly shall be unit packaged together in a fiberboard box conforming to Specification JAN-P-108. Packaging materials and fiberboard box shall conform with the general requirements of Specification JAN-P-100.

5.2 Packing

5.2.1 For Domestic Shipment - Azimuth Indicators packaged as described in paragraph 5.1 shall be packed in a shipping container conforming with the requirements of one of the following Specifications:

NN-B-601 - Style A or B, Wood-Cleated Plywood Box

NN-B-621 - Style 2 or 4, Nailed Wood Box

NN-B-631 - Wirebound Box

5.2.2 For Overseas Shipment - Azimuth Indicators packaged as described in paragraph 5.1 shall be packed in a shipping container conforming with the requirements of either Specification JAN-P-106 (style 2 or 4) or Specification JAN-P-105 (style A or B). Each box shall contain a water resistant bag or case liner (in which the contents are packed) fabricated from material conforming with the applicable requirements of Specification JAN-P-125. Closures shall be completely sealed with a continuous seam, at least 3/4 inch wide, of water resistant adhesive complying with Specification JAN-P-140.

5.2.3 The contents of both domestic and export shipping containers shall be immobilized. The gross weight shall not exceed 200 pounds for domestic shipments or 70 pounds for export shipments.

5.3 Marking - In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of Specification 100-2.

5.3.1 Each complete unit package shall be identified by marking or printing the proper data on the package or by attaching a label to the package.

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5.4 Spare Parts - Packaging, packing and marking of spare parts shall be in accordance with Drawing B7633839 and supplementary instructions supplied by the contracting officer.

6. NOTES

6.1 Ordering Data - Invitation for bids and contracts should specify the title, number, date of this specification and type of packing.

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